Discrete time Fourier Transform (DTFT)

$$\text{DTFT} \quad X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n} \quad ; \quad -\infty < \omega < \infty$$

$$\text{Inverse DTFT} \quad x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega)\, e^{j\omega n}\, d\omega$$

$X(\omega)$ is a periodic spectrum

$$X(\omega + 2\pi) = X(\omega) \quad \text{for any } \omega$$

Relation between DTFT and continuous time Fourier Series (CTFS)

CTFS

$f(t)$ be a continuous time (CT) periodic signal

fundamental period $= T_0$

$$f(t) = f(t + T_0) \quad \text{for all } t.$$

fundamental angular frequency $= \frac{2\pi}{T_0} = \Omega_0$

n^{th} harmonic $= n\Omega_0 = n\frac{2\pi}{T_0}$

$$f(t) = \sum_{n=-\infty}^{\infty} f(n)\, e^{-jn\Omega_0 t}$$

$$f(n) = \frac{1}{T_0} \int_{\langle T_0 \rangle} f(t)\, e^{+jn\Omega_0 t} \quad ; \quad n \in I$$

Variable t | $\omega \rightarrow$ Variable

Periodic Signal $f(t)$ | Periodic spectrum $X(\omega)$

fundamental Period T_0 | 2π fund. period

fundamental freq. $\frac{2\pi}{T_0} = \Omega_0$ | $\Omega_0 = \frac{2\pi}{2\pi} = 1$ fund. freq.

n^{th} harmonic $\frac{n 2\pi}{T_0} = n\Omega_0$ | $n\Omega_0 = n \cdot 1$ n^{th} harmonic

$$f(t) = \sum_n f(n)\, e^{-jn\Omega_0 t}$$

$$f(n) = \frac{1}{T_0} \int_{\langle T_0 \rangle} f(t)\, e^{jn\Omega_0 t}$$

$$X(\omega) = \sum_n x(n)\, e^{-jn\omega}$$

$$x(n) = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(\omega)\, e^{jn\omega}$$

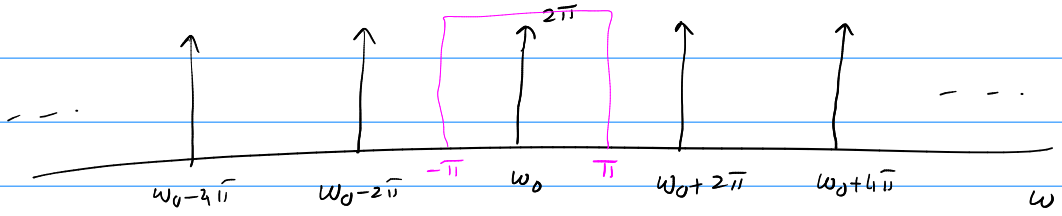
More Examples of DTFT

1. $x(n) = e^{j\omega_0 n}$, $n \in Z$

$\Downarrow$

everlasting complex exponential

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi\, \delta(\omega - (\omega_0 + k2\pi))$$



$$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\, e^{j\omega n}\, d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi} 2\pi\,\delta(\omega-\omega_0)\, e^{j\omega n}\, d\omega$$

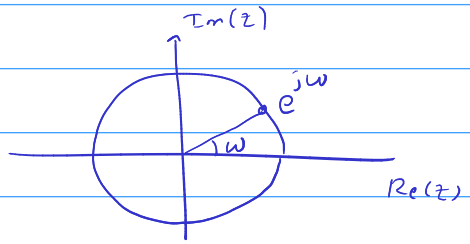
$$= e^{j\omega_0 n} = x(n)$$



Properties of DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

$$= X(z)\Big|_{@\, z = e^{j\omega}}$$ when unit circle is in ROC.



1. Linearity

$$\{x_1(n)\} \xleftrightarrow{F} X_1(\omega)$$

$$\{x_2(n)\} \xleftrightarrow{F} X_2(\omega)$$

$$\{a\,x_1(n) + b\,x_2(n)\} \xleftrightarrow{F} a\,X_1(\omega) + b\,X_2(\omega)$$

2. Time shifting

$$\{x(n)\} \xleftrightarrow{F} X(\omega)$$

$x_1(n) = x(n - N_0)$; $n \in Z$ $N_0 \rightarrow$ time shift

$$\{x_1(n)\} \xleftrightarrow{F} e^{-j\omega N_0}\, X(\omega) = X_1(\omega)$$

$$|X_1(\omega)| = |X(\omega)|$$

linear phase shift $e^{-j\omega N_0}$

phase shift is linear in ω

linear in N_0.

$e^{j\omega n} \xrightarrow[\text{Time shift } N_0]{} e^{j\omega(n-N_0)} = e^{j\omega n} \cdot e^{-j\omega N_0}$

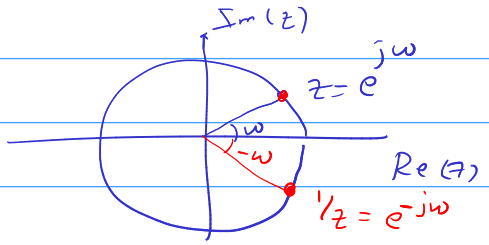
3. Time Reversal

z trans

$\{x(n)\} \xleftrightarrow{F} X(\omega) \qquad X(z)$

$x_1(n) = x(-n) \qquad X_1(z) = X(1/z)$

$\{x_1(n)\} \xleftrightarrow{F} X_1(\omega) = X(-\omega)$ (Frequency reversal)



4. Conjugation

$\{x(n)\} \xleftrightarrow{F} X(\omega) \qquad X(z)$

$x_1(n) = x^*(n) \qquad X_1(z) = X^*(z^*)$

$= (X(z^*))^*$

$\{x_1(n)\} \xleftrightarrow{F} X_1(\omega) = X^*(-\omega)$

$\{x^*(n)\} \xleftrightarrow{F} X^*(-\omega)$

5. Symmetry

- $\{x(n)\} \xleftrightarrow{F} X(\omega) = |X(\omega)|\, e^{j\angle X(\omega)}$

(Magnitude, phase)

$X(\omega) = X_R(\omega) + j X_I(\omega)$

$X_R(\omega)$: real part; $X_I(\omega)$: Imag. Part

(A) $\{x(n)\}$ is real valued

$$x(n) = x^*(n)$$

F F

$$X(w) = X^*(-w)$$ Conjugate Symmetry

$$X(-w) = X_R(-w) + jX_I(-w)$$

$$X_R(w) + jX_I(w) = X_R(-w) - jX_I(-w)$$

$$X^*(-w)$$

$$X_R(w) = X_R(-w) \quad \text{(even function)}$$

$$X_I(w) = -X_I(-w) \quad \text{(odd function)}$$

$$|X(w)| = |X(-w)| \quad \text{(even function)}$$

$$\angle X(w) = -\angle X(-w) \quad \text{(odd function)}$$

(B) $\{x(n)\}$ is real and even

$$x^*(n) = x(n) = x(-n)$$

DTFT DTFT

$$X^*(-w) = X(w) = X(-w) = X^*(-w)$$

Even Spectrum Spectrum is real

$X(w)$ is real and even.

(C) $\{x(n)\}$ is real and odd

$$-x(-n) = x(n) = x^*(n)$$

$X(\omega)$ is purely imaginary & odd. (Verify)

(D) $\{x(n)\}$ is an arbitrary signal

with spectrum $X(\omega) = X_R(\omega) + j X_I(\omega)$

$$x_c(n) = \frac{x(n) + x^*(-n)}{2}$$

$$x_a(n) = \frac{x(n) - x^*(-n)}{2}$$

$$x(n) = x_c(n) + x_a(n)$$

$$x_c(n) \overset{\mathcal{F}}{\longleftrightarrow} X_R(\omega)$$

$$x_a(n) \overset{\mathcal{F}}{\longleftrightarrow} j X_I(\omega)$$

(Verify)

6. Modulation property

$$\{x(n)\} \overset{\mathcal{F}}{\longleftrightarrow} X(\omega)$$

$$\{x(n) e^{j\omega_0 n}\} \longleftrightarrow X(\omega - \omega_0)$$

linear phase in time

Frequency shift (dual of time shift)

7.

7. Differentiation

$$\{x(n)\} \xleftrightarrow{F} X(w)$$

$$\{n\,x(n)\} \longleftrightarrow j\frac{d}{dw}X(w)$$

8. Convolution property

input $\{x(n)\} \xleftrightarrow{F} X(w)$

impulse resp $\{h(n)\} \longrightarrow H(w)$ $\Rightarrow$ called as frequency response

output $\{y(n)\} \xrightarrow{F} Y(w) = X(w) \cdot H(w)$

$\|$

$$\{x(n)\} * \{h(n)\} \xleftrightarrow{F} X(w)\,H(w)$$

Frequency response $H(w)$ $\Rightarrow$ Gain given by system for frequency w.

$\Downarrow$

Another way to characterize LTI systems.

Convolution in time domain $\Longleftrightarrow$ Multiplication in frequency domain

Q. Windowing Property

$\{x(n)\} \xleftrightarrow{F} X(\omega)$

$\{w(n)\} \xleftrightarrow{F} W(\omega)$

$\{x(n) \cdot w(n) = v(n)\} \xleftrightarrow{F} V(\omega) = \frac{1}{2\pi} \int_{\theta=-\pi}^{\pi} X(\theta)\, W(\omega-\theta)\, d\theta$

Multiplication in time domain

Periodic convolution in freq. domain